Module B6 — Field and Particle Physics

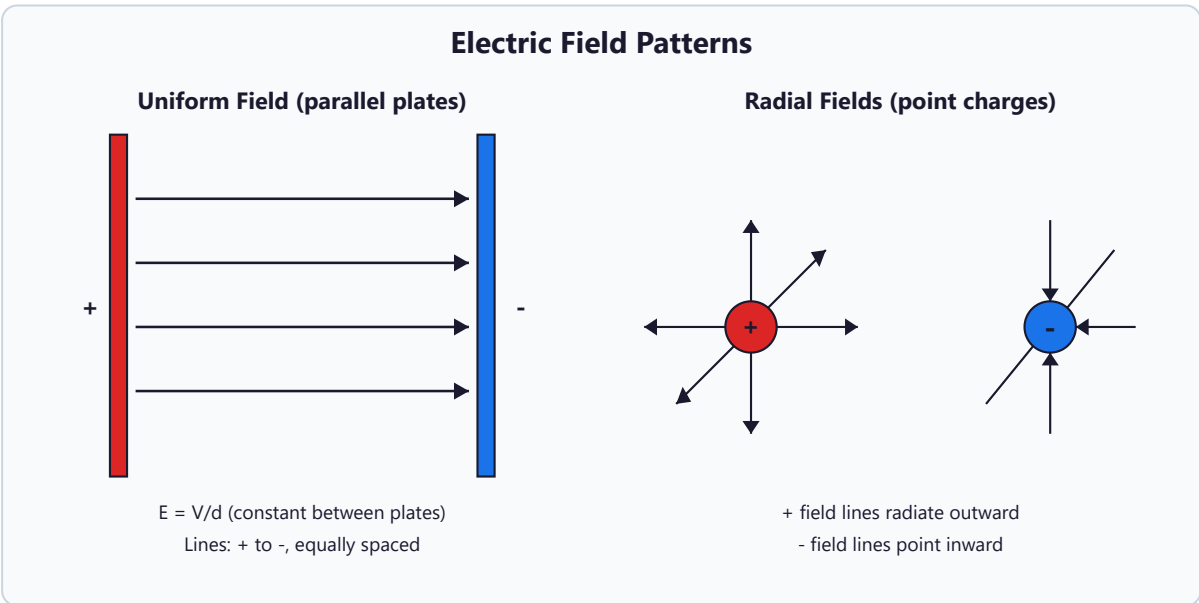
6.1 Electric Fields

Electric field: A region of space where a charged particle experiences a force. The electric field strength at a point is the force per unit positive charge placed at that point: E = F/Q.

Coulomb's law: The force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$F = kQ_1Q_2 / r^2 = Q_1Q_2 / (4\pi\varepsilon_0 r^2)$$

where $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9$ N m² C^{-2} and ε_0 is the permittivity of free space (8.85×10^{-12} F m^{-1}).



Radial Electric Fields

The electric field around a point charge or charged sphere is radial (field lines point radially outward from a positive charge, inward toward a negative charge).

$$E = kQ / r^2 = Q / (4\pi\varepsilon_0 r^2)$$

Uniform Electric Fields

Between two parallel charged plates with a potential difference V and separation d, the electric field is uniform (the same magnitude and direction everywhere between the plates).

$$E = V / d$$

A charged particle in a uniform electric field experiences a constant force and undergoes uniform acceleration, analogous to projectile motion in a gravitational field.

Electric Potential

Electric potential (V): The work done per unit positive charge in bringing a small positive test charge from infinity to that point. For a point charge: V = kQ/r. Electric potential is a scalar quantity.

$$V = kQ / r = Q / (4\pi\varepsilon_0 r)$$

$$\text{Electric potential energy: } E_p = kQ_1Q_2 / r$$

The potential is positive around a positive charge and negative around a negative charge. The work done in moving a charge q through a potential difference ΔV is W = qΔV.

Exam Tip: Electric and gravitational fields have many parallels. Coulomb's law has the same form as Newton's law of gravitation. However, electric fields can be attractive or repulsive (since charges can be positive or negative), while gravitational fields are always attractive. In exams, you may be asked to compare and contrast the two types of field.

6.2 Capacitors

Capacitance (C): The charge stored per unit potential difference across the capacitor. C = Q/V. Measured in farads (F), where 1 F = 1 C V^{-1}. In practice, capacitances are usually measured in μF, nF, or pF.

$$C = Q / V$$

$$C = \varepsilon_0 \varepsilon_r A / d \quad \text{(parallel plate capacitor)}$$

where A is the area of overlap of the plates, d is the separation, and ε_r is the relative permittivity (dielectric constant) of the material between the plates.

Energy Stored in a Capacitor

$$E = ½QV = ½CV^2 = ½Q^2/C$$

The energy stored equals the area under the charge-voltage graph (which is a straight line through the origin for a capacitor with constant capacitance).

Capacitor Combinations

Arrangement	Rule
Parallel	$C_{total} = C_1 + C_2 + C_3 + \ldots$
Series	$1/C_{total} = 1/C_1 + 1/C_2 + 1/C_3 + \ldots$

Note that capacitors combine in the opposite way to resistors: capacitors in parallel add directly, while capacitors in series add reciprocally.

Charging and Discharging

When a capacitor discharges through a resistor, the charge, current, and voltage all decay exponentially:

$$Q = Q_0 e^{-t/RC}$$

$$V = V_0 e^{-t/RC}$$

$$I = I_0 e^{-t/RC}$$

Time constant (τ): The product RC, measured in seconds. It is the time taken for the charge (or voltage or current) to fall to 1/e (about 37%) of its initial value. After 5 time constants, the capacitor is considered fully discharged (less than 1% of original charge remains).

During charging through a resistor, the charge and voltage increase exponentially toward their maximum values:

$$Q = Q_{max}(1 - e^{-t/RC})$$

$$V = V_0(1 - e^{-t/RC})$$

Exam Tip: To find the time constant from a discharge graph, read off the time at which the value falls to 37% of the initial value. Alternatively, plot ln(V) against t — the gradient gives −1/RC. This linearisation technique is frequently tested in OCR B.

6.3 Magnetic Fields

Magnetic flux density (B): A measure of the strength of a magnetic field. It is defined as the force per unit length per unit current on a straight conductor carrying current at right angles to the field. Measured in tesla (T).

Force on a Current-Carrying Conductor

$$F = BIl \sin \theta$$

where B is the magnetic flux density, I is the current, l is the length of conductor in the field, and θ is the angle between the current direction and the magnetic field. The direction of the force is given by Fleming's left-hand rule.

Force on a Moving Charge

$$F = BQv \sin \theta$$

where Q is the charge and v is its velocity. For a charge moving perpendicular to the field (θ = 90°), the magnetic force is always perpendicular to the velocity, so it acts as a centripetal force causing circular motion:

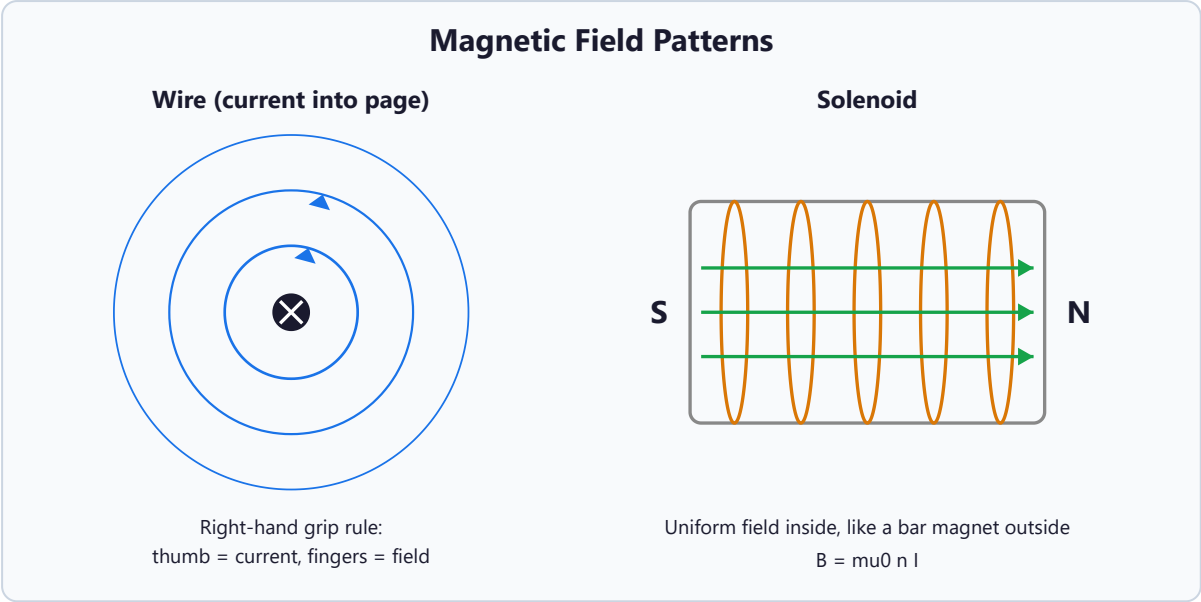
$$BQv = mv^2/r \quad \text{giving} \quad r = mv \,/\, (BQ)$$

The radius of the circular path depends on the momentum of the particle and the magnetic flux density. This principle is used in mass spectrometers and particle accelerators.

Magnetic Field Patterns

- **Long straight wire:** Concentric circles centred on the wire. Direction given by the right-hand grip rule.
- **Solenoid:** Uniform field inside (parallel field lines), similar to a bar magnet outside. Field strength inside: $B = \mu_0 nI$, where n is the number of turns per unit length.

$$B = \mu_0 nI \quad \text{(inside a solenoid)}$$



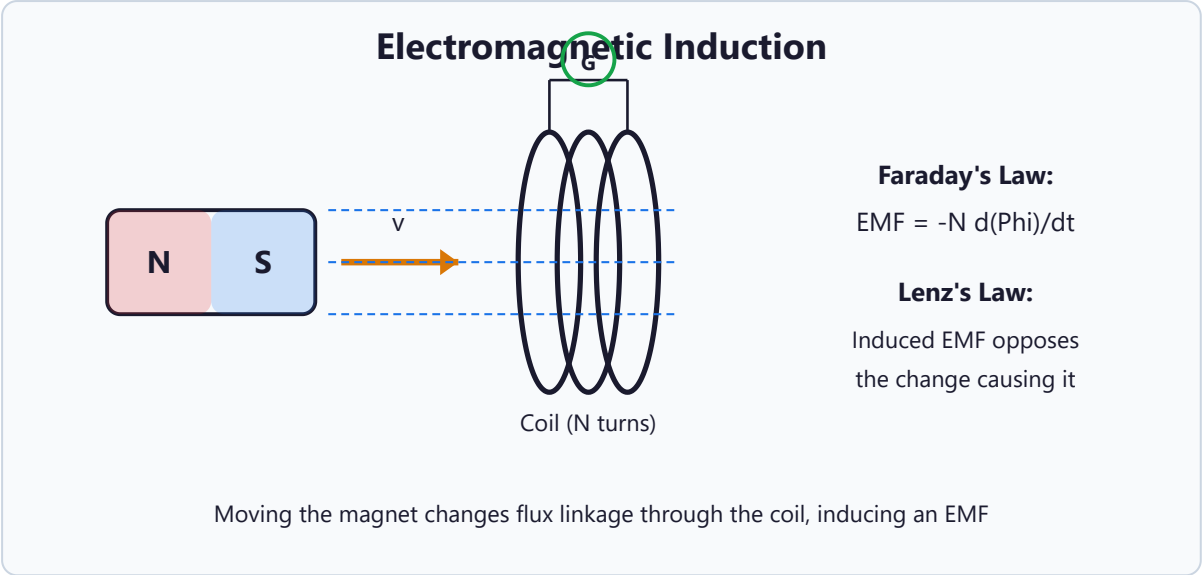
where μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ T m A^{-1}).

6.4 Electromagnetic Induction

Magnetic flux (Φ): The product of the magnetic flux density and the area perpendicular to the field. $\Phi = BA \cos \theta$. Measured in webers (Wb), where 1 Wb = 1 T m^2.

Magnetic flux linkage: For a coil with N turns, the flux linkage is $N\Phi = BAN \cos \theta$.

$$\Phi = BA \cos \theta$$



Faraday's Law

Faraday's law: The magnitude of the induced EMF is equal to the rate of change of magnetic flux linkage through the circuit.

$$\varepsilon = -N \times d\Phi/dt = -d(N\Phi)/dt$$

Lenz's Law

Lenz's law: The direction of the induced EMF (and hence the induced current) is such that it opposes the change producing it. This is a consequence of conservation of energy — the negative sign in Faraday's law.

Applications of electromagnetic induction include generators (AC and DC), transformers, and eddy current braking.

Transformers

$$V_s / V_p = N_s / N_p$$

For an ideal transformer (100% efficient): $V_p I_p = V_s I_s$. Step-up transformers increase voltage ($N_s > N_p$), step-down transformers decrease it. Transformers only work with alternating current because a changing

current is needed to produce a changing magnetic flux.

Exam Tip: Lenz's law questions often ask for the direction of induced current. Think about what magnetic field the induced current would produce, and check that it opposes the change. If flux is increasing through a coil, the induced current creates a field that opposes the increase (pointing in the opposite direction to the existing field).

6.5 Nuclear Physics

Radioactive Decay

Radioactive decay: The spontaneous and random emission of radiation from an unstable nucleus. "Spontaneous" means it is not triggered by external conditions. "Random" means it is impossible to predict which nucleus will decay next or when.

Types of Radiation

Property	Alpha (α)	Beta-minus (β^-)	Beta-plus (β^+)	Gamma (γ)
Particle	Helium nucleus (^4_2He)	Electron (e^-)	Positron (e^+)	High-energy photon
Charge	+2e	−e	+e	0
Mass	4u	~0 (1/1836 u)	~0 (1/1836 u)	0
Penetration	Few cm in air; stopped by paper	~1 m in air; stopped by few mm aluminium	~1 m in air; annihilates with electron	Very penetrating; reduced by thick lead
Ionising ability	Strongly ionising	Moderately ionising	Moderately ionising	Weakly ionising

Deflection in fields	Deflected (low charge-to-mass ratio)	Deflected strongly (opposite to α)	Deflected strongly (same as α)	Not deflected

Decay Equations

In all nuclear decay equations, the total nucleon number (mass number, A) and total proton number (atomic number, Z) must be conserved.

Alpha decay:

$$^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

Beta-minus decay: A neutron transforms into a proton, emitting an electron and an antineutrino.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$$

Beta-plus decay: A proton transforms into a neutron, emitting a positron and a neutrino.

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}\beta + \nu_e$$

Exponential Decay and Half-Life

Half-life ($t_{½}$): The average time taken for half of the undecayed nuclei in a sample to decay, or equivalently, the time for the activity or count rate to halve.

$$N = N_0 e^{-\lambda t}$$

$$A = A_0 e^{-\lambda t} = \lambda N$$

$$t_{½} = \ln 2 / \lambda = 0.693 / \lambda$$

where λ is the decay constant (the probability of a nucleus decaying per unit time), N is the number of undecayed nuclei, and A is the activity (number of decays per second, measured in becquerels, Bq).

Nuclear Fission and Fusion

Nuclear fission: The splitting of a heavy nucleus (e.g., uranium-235) into two lighter nuclei, typically triggered by the absorption of a neutron. The products have a lower total mass than the original nucleus, with the mass difference released as energy ($E = mc^2$). Fission also releases neutrons, which can cause further fission in a chain reaction.

Nuclear fusion: The combining of two light nuclei to form a heavier nucleus. Again, the products have lower total mass and the difference is released as energy. Fusion requires extremely high temperatures (millions of kelvin) to overcome the electrostatic repulsion between the positively charged nuclei.

Binding Energy

Mass defect (Δm): The difference between the mass of the separated nucleons and the mass of the assembled nucleus. The mass of the nucleus is always less than the sum of its individual nucleons.

Binding energy: The energy equivalent of the mass defect. It is the minimum energy required to completely separate a nucleus into its individual protons and neutrons. Binding energy = $\Delta m \times c^2$.

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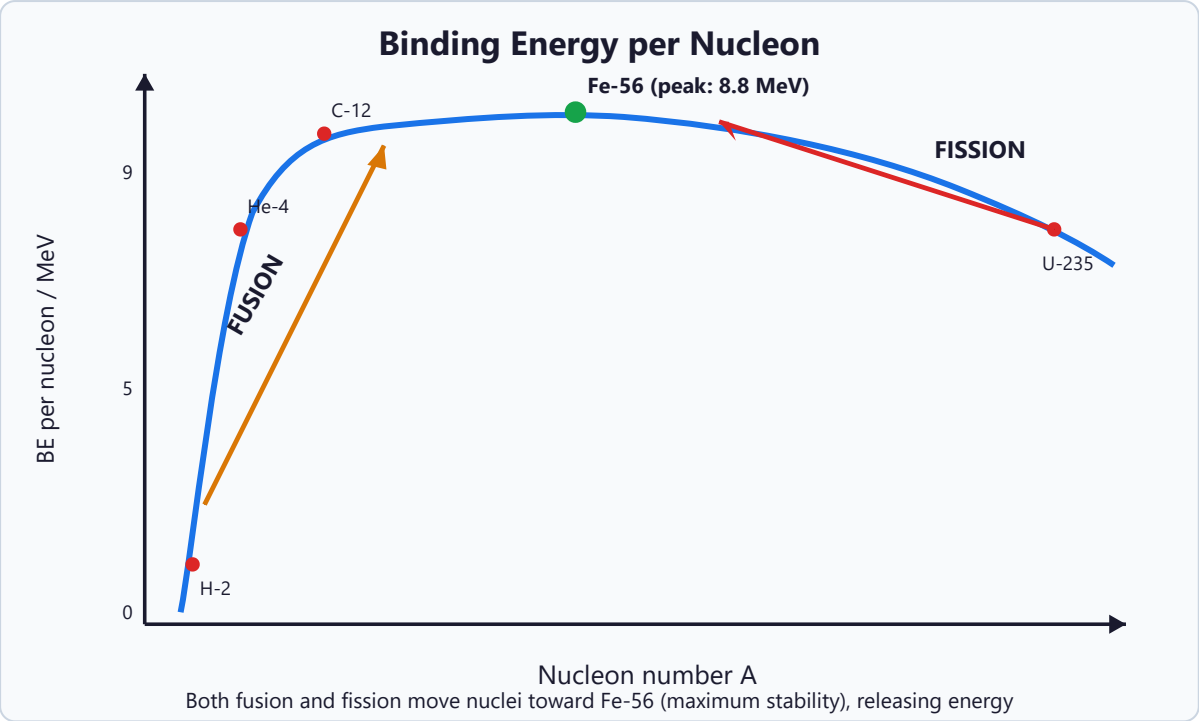
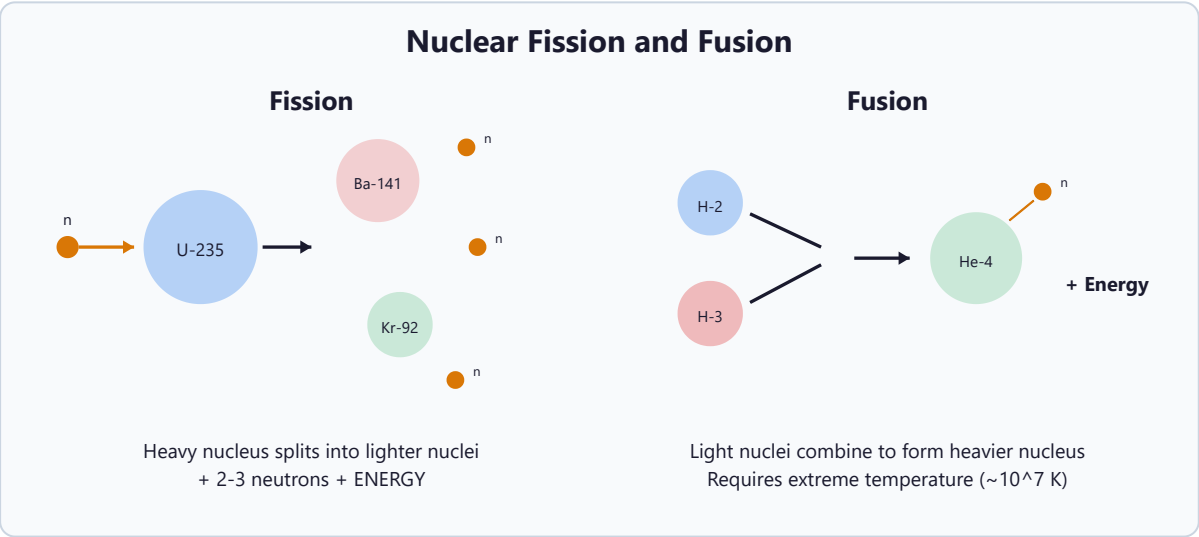




The binding energy per nucleon curve peaks at iron-56 (about 8.8 MeV per nucleon). This means:

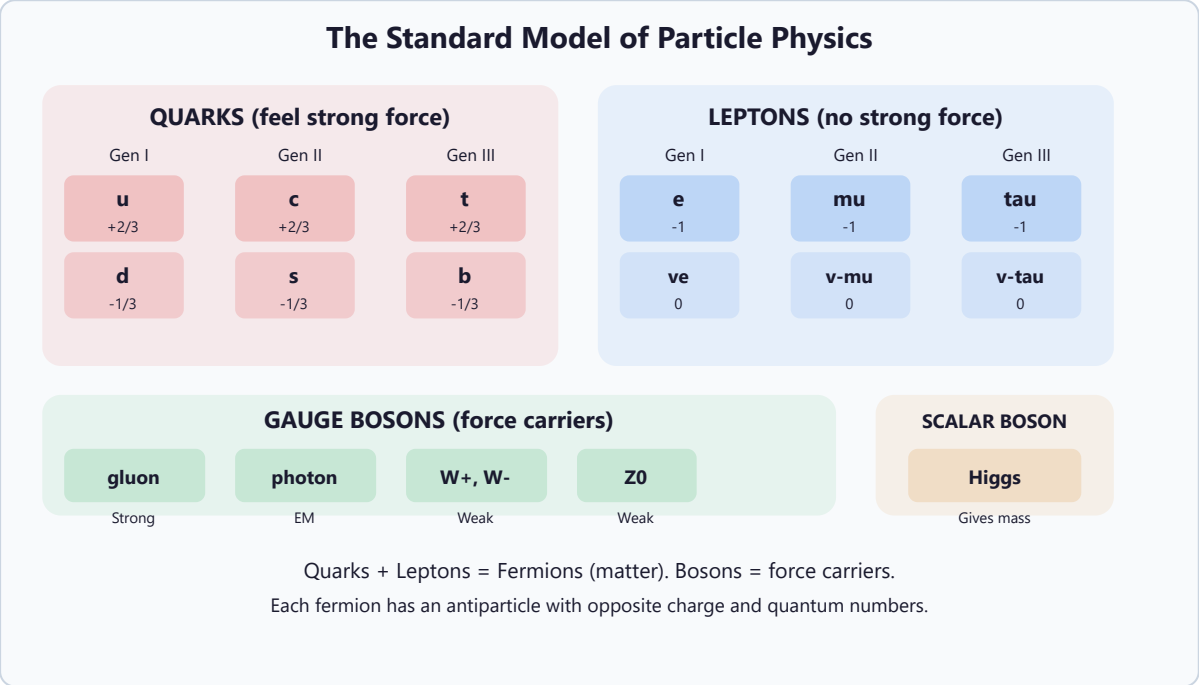
- Nuclei lighter than iron can release energy by fusion (moving up the curve)
- Nuclei heavier than iron can release energy by fission (also moving toward the peak)

Exam Tip: A higher binding energy per nucleon means a more stable nucleus. Energy is released in nuclear reactions when the products are more tightly bound (higher binding energy per nucleon) than the reactants. The energy released equals the increase in total binding energy.

6.6 Fundamental Particles

The Standard Model

The Standard Model classifies all known fundamental particles into two groups: fermions (matter particles) and bosons (force carriers).



Quarks

Quarks are fundamental particles that combine to form hadrons. There are six flavours of quark:

Quark	Symbol	Charge (e)	Baryon Number
Up	u	+2/3	+1/3
Down	d	−1/3	+1/3
Charm	c	+2/3	+1/3
Strange	s	−1/3	+1/3
Top	t	+2/3	+1/3

Bottom	b	−1/3	+1/3

Each quark has a corresponding antiquark with opposite charge and baryon number.

Hadrons: Baryons and Mesons

Hadron: A particle made of quarks, held together by the strong force. Hadrons are either baryons or mesons.

Baryon: A hadron made of three quarks (or three antiquarks for an antibaryon). Examples: proton (uud), neutron (udd).

Meson: A hadron made of a quark-antiquark pair. Examples: pion (π), kaon (K).

Particle	Quark Composition	Charge	Baryon Number
Proton (p)	uud	+1	+1
Neutron (n)	udd	0	+1
Antiproton	$\bar{u}\bar{u}\bar{d}$	−1	−1
π^+	$u\bar{d}$	+1	0
π^-	$\bar{u}d$	−1	0
π^0	$u\bar{u}$ or $d\bar{d}$	0	0
K^+	$u\bar{s}$	+1	0

Leptons

Leptons are fundamental particles that do not experience the strong force. There are six leptons:

- Electron (e^-) and electron neutrino (ν_e)
- Muon (μ^-) and muon neutrino (ν_μ)

- Tau (τ^-) and tau neutrino (ν_τ)

Each lepton has a corresponding antiparticle. Lepton number is conserved in all interactions (electron lepton number and muon lepton number are conserved separately).

Exchange Particles (Gauge Bosons)

Forces between particles are mediated by the exchange of gauge bosons:

Force	Gauge Boson	Acts On	Range
Strong nuclear	Gluon (g)	Quarks (and gluons)	$\sim 10^{-15}$ m
Electromagnetic	Photon (γ)	Charged particles	Infinite
Weak nuclear	W^+, W^-, Z^0	All fermions	$\sim 10^{-18}$ m
Gravity	Graviton (hypothetical)	All particles with mass	Infinite

Conservation Laws in Particle Interactions

In every particle interaction, the following quantities must be conserved:

- **Charge** — always conserved
- **Baryon number** — always conserved
- **Lepton number** — conserved separately for each generation (electron, muon, tau)
- **Strangeness** — conserved in strong and electromagnetic interactions, can change by ±1 in weak interactions
- **Energy and momentum** — always conserved

Exam Tip: When checking whether a particle interaction is possible, apply all conservation laws. If any one is violated, the interaction cannot occur (or can only occur via a specific force). The weak force is the only force that can change quark flavour, which is why it is responsible for beta decay and strangeness-changing interactions.

Beta Decay at the Quark Level

Beta-minus decay involves a down quark changing into an up quark, mediated by a W^- boson:

$$d \rightarrow u + W^- \quad \text{then} \quad W^- \rightarrow e^- + \bar{\nu}_e$$

Beta-plus decay involves an up quark changing into a down quark, mediated by a W^+ boson:

$$u \rightarrow d + W^+ \quad \text{then} \quad W^+ \rightarrow e^+ + \nu_e$$

Annihilation and Pair Production

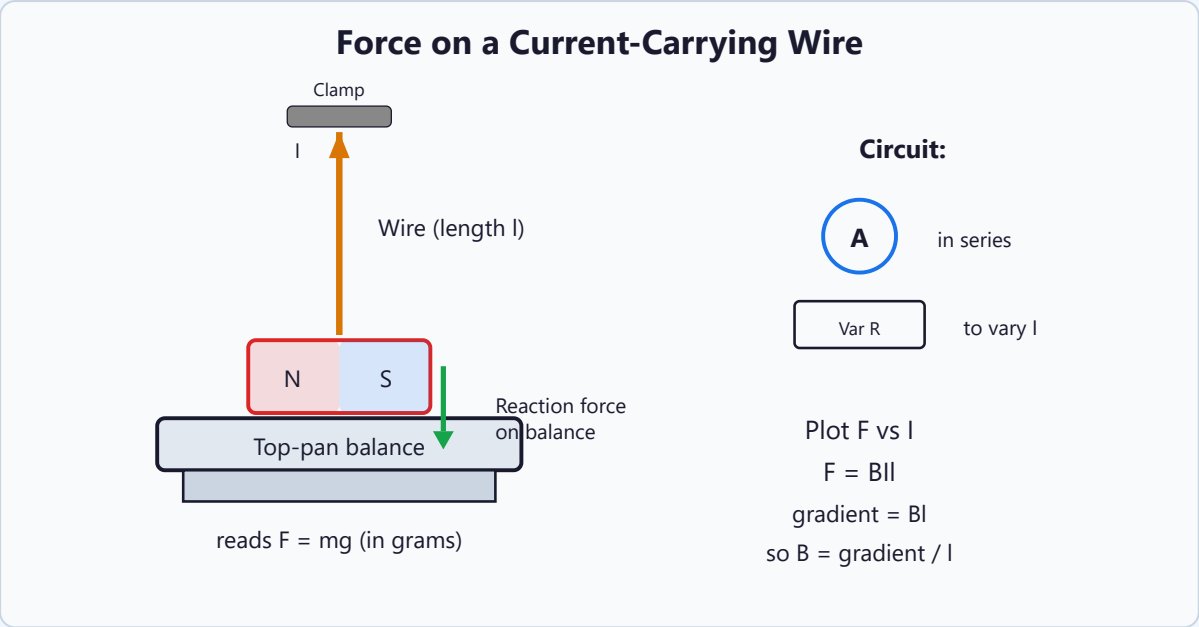
Annihilation: When a particle meets its antiparticle, they annihilate each other, converting their mass into energy in the form of two gamma photons (to conserve momentum).

Pair production: A high-energy photon can spontaneously create a particle-antiparticle pair (usually near a nucleus to conserve momentum). The photon must have at least enough energy to create the rest masses of both particles: $E_{min} = 2mc^2$.

$$\text{Minimum photon energy for pair production: } E = 2m_0c^2$$

Exam Tip: In annihilation, each gamma photon has energy equal to the rest mass energy of one particle ($E = m_0c^2$), assuming the particles were approximately at rest. For an electron-positron annihilation, each photon has energy 0.511 MeV. Two photons are produced travelling in opposite directions to conserve momentum.

Practical Activity: Force on a Current-Carrying Wire



Apparatus

- Strong U-shaped magnets (or slab magnets with yoke)
- Stiff wire (straight section of known length l) clamped above the magnets
- Top-pan balance (sensitive to 0.01 g) with the magnets placed on it
- DC power supply, ammeter (in series), variable resistor
- Connecting leads, switch

Method

1. Place the magnets on the top-pan balance. Zero the balance.
2. Position the wire so it passes horizontally between the poles of the magnet, perpendicular to the field.
3. Set the current to a low value using the variable resistor. Record I from the ammeter and the balance reading (in grams).
4. The force on the wire (by Newton's third law) produces an equal and opposite reaction on the magnets, changing the balance reading.
5. Convert the balance reading to force: F = (change in mass reading) x g / 1000.
6. Increase the current in steps and record F and I each time (at least 6 values).
7. Plot F (y-axis) against I (x-axis). The gradient = Bl, so B = gradient / l.

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Safety

- Keep current low to avoid overheating the wire
- Strong magnets can pinch fingers — handle carefully
- Keep magnetic-sensitive equipment (phones, cards) away from magnets

Analysis

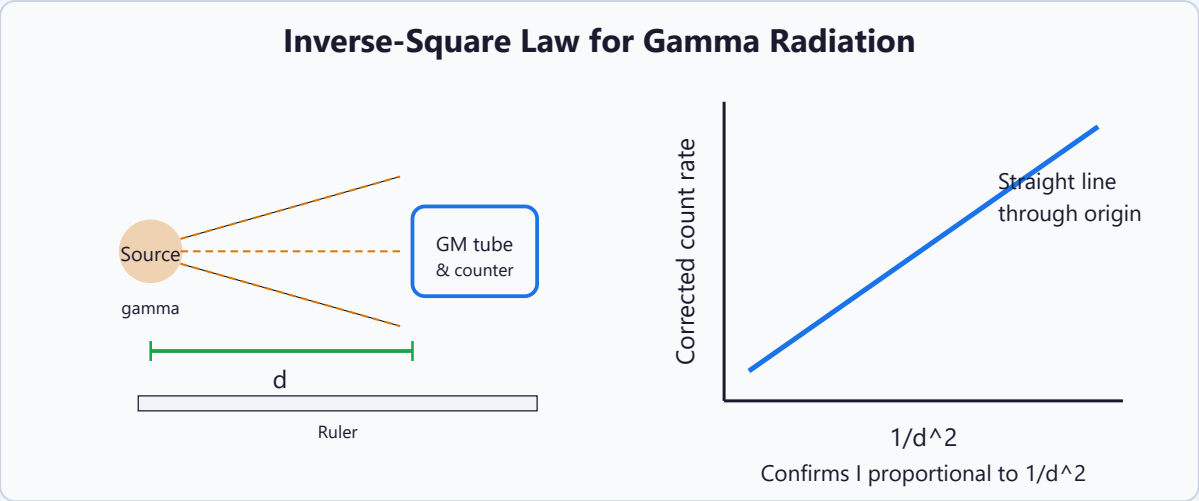
From $F = BIl$, a graph of F against I should be a straight line through the origin with gradient = Bl. Dividing by the measured length l of wire in the field gives the magnetic flux density B.

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire not exactly perpendicular to the field	Systematic (F too small)	Use a rigid wire clamped carefully; check alignment
Non-uniform magnetic field at the edges	Systematic	Use slab magnets with a yoke for a more uniform field; keep wire in the centre
Balance sensitivity	Random	Use a balance with 0.001 g resolution; shield from draughts

Exam Tip: The balance measures the reaction force to the force on the wire (Newton's third law). If the wire is pushed up, the magnets are pushed down, increasing the balance reading. If the current is reversed, the force direction reverses and the balance reading decreases.

Practical Activity: Inverse-Square Law for Gamma Radiation



Apparatus

- Sealed gamma source (e.g., Co-60 or Cs-137) in a holder
- Geiger-Muller (GM) tube and counter/ratemeter
- Metre ruler (to measure source-to-detector distance d)
- Stopwatch (for timed counts)

Method

1. First, measure the background count rate (with no source present) over at least 5 minutes. Calculate the mean background count rate.
2. Place the gamma source at a known distance d from the GM tube window.
3. Record the count over a fixed time (e.g., 60 seconds). Repeat 3 times and find the mean count rate.
4. Subtract the background count rate to get the corrected count rate.
5. Increase d in steps (e.g., 5 cm to 50 cm in 5 cm steps) and repeat.
6. Plot corrected count rate (y-axis) against 1/d^2 (x-axis). A straight line through the origin confirms the inverse-square law.

Safety

- Follow the ALARA principle: As Low As Reasonably Achievable exposure
- Handle the source with long tongs, never with bare hands
- Point the source away from people; keep exposure time minimal

- Store the source in a lead-lined container when not in use
- Wear a radiation monitoring badge

Analysis

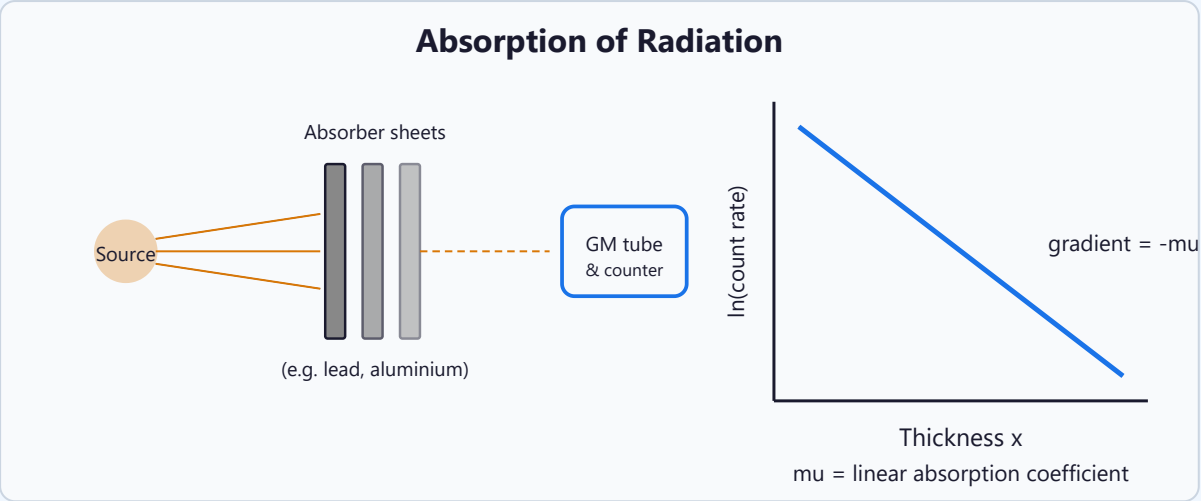
The intensity of gamma radiation follows I proportional to 1/d^2 (inverse-square law). The corrected count rate is proportional to intensity, so plotting corrected count rate against 1/d^2 should give a straight line through the origin.

Sources of Error & Improvements

Source of Error	Type	Improvement
Random nature of radioactive decay causes count rate to vary	Random	Count for longer periods (e.g., 5 minutes); take repeat readings
Measuring d from the wrong point (not the centre of the source)	Systematic	Measure from source to GM tube window consistently; plot corrected count vs 1/(d+d0)^2 and adjust d0
Background radiation not properly subtracted	Systematic	Measure background multiple times and take a reliable mean
Dead time of GM tube at high count rates (close distances)	Systematic	Avoid very close distances; use a weaker source if count rates exceed ~500 s^-1

Exam Tip: Always subtract background radiation from your readings. If the graph does not pass through the origin, it may be because the effective centre of the source is not at the measured reference point. You can correct for this by using d + d0 instead of d.

Practical Activity: Absorption of Radiation



Apparatus

- Gamma source (e.g., Co-60 or Cs-137) or beta source
- Geiger-Muller (GM) tube and counter
- Absorber sheets of known thickness (e.g., lead sheets of varying thickness, or aluminium for beta)
- Ruler or micrometer (to measure absorber thickness)
- Stopwatch

Method

1. Measure the background count rate (no source, no absorber) over at least 5 minutes.
2. Place the source a fixed distance from the GM tube. Record the count rate with no absorber.
3. Insert one sheet of absorber between the source and GM tube. Record the count rate.
4. Add more sheets progressively, recording the corrected count rate (minus background) and the total absorber thickness each time.
5. Plot ln(corrected count rate) (y-axis) against absorber thickness x (x-axis).
6. A straight line with negative gradient confirms exponential absorption: I = I0 e^(-mu x).

Safety

- Follow all radioactive source safety procedures (tongs, minimal exposure time, lead storage)
- Handle lead absorbers with gloves (lead is toxic)
- Wash hands after the experiment

Analysis

From I = I0 e^(-mu x), taking natural logs gives ln(I) = ln(I0) - mu x. The gradient of ln(count rate) vs thickness gives -mu (the linear absorption coefficient). The half-value thickness x_half = ln(2) / mu.

Sources of Error & Improvements

Source of Error	Type	Improvement
Statistical variation in count rate	Random	Count for longer time periods; repeat and average
Absorber thickness not uniform	Systematic	Measure thickness at several points with micrometer; use machined sheets
Scattered radiation reaching detector	Systematic	Use a collimated beam; place absorber close to the source

Exam Tip: The half-value thickness is the thickness of absorber that reduces the count rate to half. It is related to the absorption coefficient by x_half = ln(2) / mu. This is analogous to radioactive half-life but for attenuation of radiation through matter.

Worked Example: Two point charges of +3.0 μC and −5.0 μC are separated by 0.20 m. Calculate the force between them and state whether it is attractive or repulsive.

$F = kQ_1Q_2/r^2 = (8.99 \times 10^9) \times (3.0 \times 10^{-6}) \times (5.0 \times 10^{-6}) / (0.20)^2$

$F = (8.99 \times 10^9 \times 1.5 \times 10^{-11}) / 0.04$

$F = 0.1349 / 0.04 =$ **3.4 N**

The charges are opposite in sign, so the force is **attractive**.

Worked Example: A proton ($m = 1.67 \times 10^{-27}$ kg, $Q = 1.60 \times 10^{-19}$ C) moves at 4.0×10^6 m s^{-1} perpendicular to a magnetic field of 0.80 T. Calculate the radius of its circular path.

$r = mv/(BQ) = (1.67 \times 10^{-27} \times 4.0 \times 10^{6}) / (0.80 \times 1.60 \times 10^{-19})$

$r = 6.68 \times 10^{-21} / 1.28 \times 10^{-19}$ = **0.052 m = 5.2 cm**

A heavier particle (e.g. an alpha particle with 4 times the mass and 2 times the charge) would have a larger radius: $r_\alpha = (4m)(v) / (B \times 2Q) = 2r_p$. This is how mass spectrometers separate ions by mass-to-charge ratio.

Worked Example: A radioactive source has an activity of 6400 Bq. After 3 hours, the activity is 800 Bq. Calculate the half-life.

$A = A_0(1/2)^n$, where n is the number of half-lives

$800 = 6400 \times (1/2)^n$

$(1/2)^n = 800/6400 = 1/8 = (1/2)^3$

So n = 3 half-lives in 3 hours

$t_{1/2}$ = 3 hours / 3 = **1.0 hour = 3600 s**

Check: $\lambda = \ln 2/t_{1/2} = 0.693/3600 = 1.93 \times 10^{-4}\ s^{-1}$

$A = 6400 \times e^{-(1.93 \times 10-4)(10800)} = 6400 \times e^{-2.079} = 6400 \times 0.125 = 800$ Bq ✓